

MACROECONOMIC THEORY I — CLASS NOTES

Lectures 1–3: The Solow Growth Model

Setup, Dynamics, Evaluation, Differential Equations, and Policy

ECON*6020, Fall 2026

How to read these notes The notes follow the four slide decks in order. Each section starts with a grey line telling you which slides it covers: **L1** = *The Solow Growth Model* (15 slides), **EV** = *Evaluating the Solow Growth Model* (18), **DE** = *Differential equations / speed of convergence* (14), **PO** = *Policy questions in the Solow model* (8).

- blue boxes — results to remember (mostly exactly as on the slides).
- green bars — the idea in plain words.
- orange bars — easy-to-make mistakes.
- Dashed boxes marked *Beyond the slides* — worked checks and extra steps that are *not* on the slides. Skip them if you only want the lecture content.
- In derivations, the grey italic text at the right edge says *what was done* on that line.

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1 Notation and the Growth-Rate Toolkit

Slides: L1 2–3

1.1 Three ways to measure every variable

The model is written three times over, once for each way of measuring a quantity:

Written as	Example	Definition	Meaning
Upper case	Y_t, K_t	—	aggregate (whole economy)
Lower case	y_t, k_t	$y_t = Y_t/L_t$	per capita (per worker)
Hat	$\hat{y}_t, \hat{k}_t$	$\hat{y}_t = Y_t/(A_t L_t)$	per <i>effective</i> unit of labour

$A_t L_t$ is the number of *effective* labour units: workers L_t weighted by how productive each one is, A_t .

Variables and parameters (L1 slide 3)

Y_t	GDP (output)	A_t	productivity (labour-augmenting technology)
C_t	consumption	x	growth rate of technology
I_t	investment	w_t	wage rate
K_t	capital stock	r_t	rental rate of capital
L_t	population / labour	n	population growth rate
s	saving rate	δ	depreciation rate

The parameters s, n, x, δ are **constant and exogenous**: the model takes them as given.

1.2 Dots and growth rates

A dot is a time derivative, $\dot{X}_t \equiv dX_t/dt$: how much X changes per unit of time. Dividing by X gives the **growth rate**, the change as a *fraction* of the current level:

$$\frac{\dot{X}_t}{X_t} = \frac{dX_t/dt}{X_t}.$$

(E.g. $\dot{L}_t/L_t = 0.01$ means population grows 1% per year.)

Why a growth rate is the time derivative of a log

$$\begin{aligned} \frac{d \ln X_t}{dt} &= \frac{d \ln X_t}{dX_t} \cdot \frac{dX_t}{dt} && \text{chain rule} \\ &= \frac{1}{X_t} \cdot \dot{X}_t = \frac{\dot{X}_t}{X_t}. && d \ln X/dX = 1/X \end{aligned}$$

This one fact turns every growth-rate calculation into “take logs, then differentiate”.

The three growth-rate rules

Take logs, then use $\ln(XY) = \ln X + \ln Y$, $\ln(X/Y) = \ln X - \ln Y$, $\ln X^a = a \ln X$, and differentiate with respect to t :

$$\begin{aligned}\frac{d \ln(XY)}{dt} &= \frac{d \ln X}{dt} + \frac{d \ln Y}{dt} \implies \frac{(\dot{XY})}{XY} = \frac{\dot{X}}{X} + \frac{\dot{Y}}{Y}, & \text{product} \\ \frac{d \ln(X/Y)}{dt} &= \frac{d \ln X}{dt} - \frac{d \ln Y}{dt} \implies \frac{(\dot{X/Y})}{X/Y} = \frac{\dot{X}}{X} - \frac{\dot{Y}}{Y}, & \text{ratio} \\ \frac{d \ln(X^a)}{dt} &= a \frac{d \ln X}{dt} \implies \frac{(\dot{X}^a)}{X^a} = a \frac{\dot{X}}{X}. & \text{power}\end{aligned}$$

Growth-rate rules

Products → **add** growth rates; ratios → **subtract** them; powers → **multiply** by the exponent.

1.3 Exogenous growth of population and technology

Population and technology grow at constant rates:

$$\dot{L}_t = nL_t \iff \frac{\dot{L}_t}{L_t} = n, \quad \frac{\dot{A}_t}{A_t} = x.$$

Solving the first one (the second is identical with x in place of n):

$$\begin{aligned}\frac{d \ln L_t}{dt} &= n & \text{growth rate} &= \text{derivative of the log} \\ \ln L_t &= \ln L_0 + nt & & \text{integrate from 0 to } t \\ L_t &= L_0 e^{nt}. & & \text{exponentiate}\end{aligned}$$

Likewise $A_t = A_0 e^{xt}$. By the product rule, effective labour grows at the *sum* of the two rates:

$$\frac{(\dot{A_t L_t})}{A_t L_t} = x + n \implies A_t L_t = A_0 L_0 e^{(n+x)t}.$$

2 Production and Factor Prices

Slides: L1 4–8

2.1 The production function

Output is produced from effective labour and capital, $Y_t = F(A_t L_t, K_t)$, with three properties:

1. **Constant returns to scale (CRS).** Double both inputs and output doubles:

$$F(A_t \lambda L_t, \lambda K_t) = \lambda F(A_t L_t, K_t) = \lambda Y_t \quad \text{for any } \lambda > 0.$$

2. **Positive but diminishing marginal products.** More of an input raises output, but by less and less:

$$F_L > 0, \quad F_K > 0, \quad F_{LL} < 0, \quad F_{KK} < 0.$$

3. Inada conditions.

$$\lim_{L_t \rightarrow 0} F_L = \infty, \quad \lim_{L_t \rightarrow \infty} F_L = 0, \quad \lim_{K_t \rightarrow 0} F_K = \infty, \quad \lim_{K_t \rightarrow \infty} F_K = 0.$$

As an input becomes scarce its marginal product becomes arbitrarily large; as it becomes abundant its marginal product goes to zero.

Intensive form: one variable is enough

Under CRS we can choose $\lambda = 1/(A_t L_t)$:

$$\hat{y}_t = \frac{Y_t}{A_t L_t} = F\left(1, \frac{K_t}{A_t L_t}\right) \equiv f(\hat{k}_t).$$

So output per effective worker depends *only* on capital per effective worker. This is why the whole model can later be reduced to one equation in $\hat{k}_t$.

2.2 Cobb–Douglas production

The slides use

$$Y_t = (A_t L_t)^{1-\alpha} K_t^\alpha, \quad 0 < \alpha < 1,$$

which satisfies all three properties. Dividing by $A_t L_t$ and by L_t gives the other two versions:

$$\begin{aligned} \hat{y}_t &= \frac{(A_t L_t)^{1-\alpha} K_t^\alpha}{A_t L_t} = (A_t L_t)^{-\alpha} K_t^\alpha = \left(\frac{K_t}{A_t L_t}\right)^\alpha = \hat{k}_t^\alpha, & \text{divide by } A_t L_t \\ y_t &= \frac{(A_t L_t)^{1-\alpha} K_t^\alpha}{L_t} = A_t^{1-\alpha} L_t^{-\alpha} K_t^\alpha = A_t^{1-\alpha} \left(\frac{K_t}{L_t}\right)^\alpha = A_t^{1-\alpha} k_t^\alpha. & \text{divide by } L_t \end{aligned}$$

2.3 Factor prices and constant income shares

Competitive firms pay each factor the value of its marginal product:

$$\begin{aligned} w_t &= \frac{\partial Y_t}{\partial L_t} = (1-\alpha) A_t^{1-\alpha} L_t^{-\alpha} K_t^\alpha = (1-\alpha) \frac{Y_t}{L_t}, & \text{differentiate w.r.t. } L_t \\ r_t &= \frac{\partial Y_t}{\partial K_t} = \alpha (A_t L_t)^{1-\alpha} K_t^{\alpha-1} = \alpha \frac{Y_t}{K_t}. & \text{differentiate w.r.t. } K_t \end{aligned}$$

In per-effective-worker terms (these become equations (5) and (6) below):

$$r_t = \alpha \hat{k}_t^{\alpha-1}, \quad w_t = (1-\alpha) A_t \hat{k}_t^\alpha.$$

Multiplying w_t by L_t and r_t by K_t gives the income shares:

$$\frac{w_t L_t}{Y_t} = 1 - \alpha, \quad \frac{r_t K_t}{Y_t} = \alpha, \quad \text{so } w_t L_t + r_t K_t = Y_t.$$

Constant factor shares

Labour's share of income is $1 - \alpha$ and capital's share is α , whatever the values of K , L , A . The data show roughly this: U.S. labour's share (labour compensation / total income, annual, 1929–2024) fluctuates around 62.8% (64.2% post-1960). The standard calibration is

$$1 - \alpha = 2/3 \quad \implies \quad \alpha = 1/3.$$

3 The Equations of the Model

Slides: L1 9–11

3.1 The model in three units

	Aggregate	Per capita	Per effective worker	
Production	$Y = (AL)^{1-\alpha} K^\alpha$	$y = A^{1-\alpha} k^\alpha$	$\hat{y} = \hat{k}^\alpha$	(1)
Resource constraint	$Y = C + I$	$y = c + i$	$\hat{y} = \hat{c} + \hat{i}$	(2)
Saving = investment	$I = sY$	$i = sy$	$\hat{i} = s\hat{y}$	(3)
Capital accumulation	$\dot{K} = I - \delta K$	$\dot{k} = i - (\delta + n)k$	$\dot{\hat{k}} = \hat{i} - (\delta + n + x)\hat{k}$	(4)

plus the factor prices $r = \alpha \hat{k}^{\alpha-1}$ (5) and $w = (1 - \alpha)A\hat{k}^\alpha$ (6). Time subscripts are dropped.

Rows (1)–(3) are just the aggregate equation divided by L or by AL . Row (4) needs more care, because *the time derivative of a ratio is not the ratio of time derivatives*.

3.2 Deriving the capital-accumulation equations

Watch out $\frac{\dot{K}}{L} = \frac{\dot{I}}{L} - \delta \frac{K}{L} = i - \delta k$, but this is **not** $\dot{k}$. Capital per worker also falls when population grows, because the same K is shared among more people. Always go through growth rates.

Per capita: $k = K/L$

$$\begin{aligned}
 \frac{\dot{k}}{k} &= \frac{\dot{K}}{K} - \frac{\dot{L}}{L} && \text{ratio rule} \\
 &= \frac{I - \delta K}{K} - n && \text{use } \dot{K} = I - \delta K \text{ and } \dot{L}/L = n \\
 \dot{k} &= \frac{I - \delta K}{K} \cdot \frac{K}{L} - nk && \text{multiply both sides by } k = K/L \\
 &= \frac{I}{L} - \delta k - nk = i - (\delta + n)k. && \text{simplify}
 \end{aligned}$$

Per effective worker: $\hat{k} = K/(AL)$

$$\begin{aligned}
 \frac{\dot{\hat{k}}}{\hat{k}} &= \frac{\dot{K}}{K} - \frac{\dot{A}}{A} - \frac{\dot{L}}{L} && \text{ratio + product rules} \\
 &= \frac{I - \delta K}{K} - x - n && \text{use } \dot{K}, \dot{A}/A = x, \dot{L}/L = n \\
 \dot{\hat{k}} &= \frac{I - \delta K}{K} \cdot \frac{K}{AL} - (x + n)\hat{k} && \text{multiply by } \hat{k} = K/(AL)
 \end{aligned}$$

$$= \frac{I}{AL} - \delta \hat{k} - (x + n)\hat{k} = \hat{i} - (\delta + n + x)\hat{k}. \quad \text{simplify}$$

Intuition $(\delta + n + x)\hat{k}$ is **break-even investment**: the investment needed just to keep $\hat{k}$ constant. It must replace worn-out capital (δ), equip the new workers (n), and keep up with each worker becoming more productive (x).

3.3 The key dynamic equation

From (2) and (3), consumption is what is not saved:

$$\hat{c}_t = (1 - s)\hat{y}_t. \quad (7)$$

Substituting (1) and (3) into (4):

The Solow equation

$$\dot{\hat{k}}_t = s\hat{y}_t - (\delta + n + x)\hat{k}_t = s\hat{k}_t^\alpha - (\delta + n + x)\hat{k}_t \quad (8)$$

Actual investment minus break-even investment. It is a **nonlinear first-order differential equation** in $\hat{k}_t$. The goal (Section 8) is a solution relating $\hat{k}_t$ to time, the parameters $(s, \delta, n, x, \alpha)$, and the initial value $\hat{k}_0$.

4 Dynamics and the Steady State

Slides: L1 12–13; EV 3, 14

4.1 The Solow diagram

Plot the two terms of (8) against $\hat{k}$. Because $0 < \alpha < 1$, actual investment $s\hat{k}^\alpha$ is concave; break-even investment $(\delta + n + x)\hat{k}$ is a straight line through the origin. $\dot{\hat{k}}$ is the vertical gap between them.

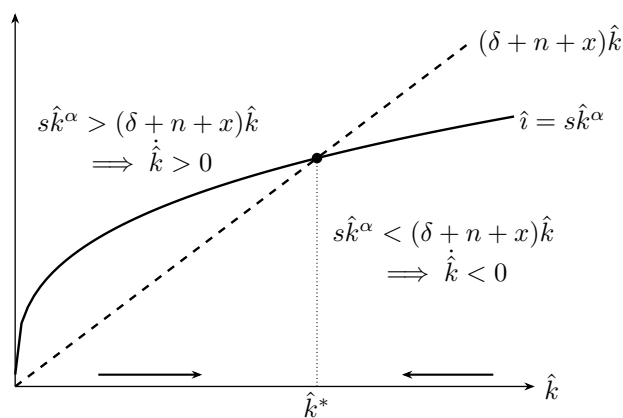


Figure 1: The Solow diagram ($s = 0.3$, $\alpha = 1/3$, $\delta + n + x = 0.08$). Arrows show the direction in which $\hat{k}$ moves.

Where $\hat{k}$ is	Comparison	Consequence
$\hat{k}_t < \hat{k}^*$	$\hat{i}_t > (\delta + n + x)\hat{k}_t$	$\dot{\hat{k}}_t > 0$: $\hat{k}$ rises
$\hat{k}_t = \hat{k}^*$	$\hat{i}_t = (\delta + n + x)\hat{k}_t$	$\dot{\hat{k}}_t = 0$: steady state
$\hat{k}_t > \hat{k}^*$	$\hat{i}_t < (\delta + n + x)\hat{k}_t$	$\dot{\hat{k}}_t < 0$: $\hat{k}$ falls

4.2 Solving for the steady state

Set $\dot{\hat{k}} = 0$ in (8) and solve for $\hat{k} > 0$:

$$\begin{aligned}
 0 &= s(\hat{k}^*)^\alpha - (\delta + n + x)\hat{k}^* && \text{steady state: } \dot{\hat{k}} = 0 \\
 (\hat{k}^*)^{\alpha-1} &= \frac{\delta + n + x}{s} && \text{divide by } s\hat{k}^* \\
 \hat{k}^* &= \left(\frac{s}{\delta + n + x} \right)^{\frac{1}{1-\alpha}}. && \text{raise to the power } \frac{1}{\alpha-1}
 \end{aligned}$$

The other steady-state values follow:

$$\hat{y}^* = (\hat{k}^*)^\alpha = \left(\frac{s}{\delta + n + x} \right)^{\frac{\alpha}{1-\alpha}}, \quad \hat{c}^* = (1-s)\hat{y}^*, \quad \left(\frac{Y}{L} \right)^* = A\hat{y}^* = A \left(\frac{s}{\delta + n + x} \right)^{\frac{\alpha}{1-\alpha}}.$$

Steady state

$$\hat{k}^* = \left(\frac{s}{\delta + n + x} \right)^{\frac{1}{1-\alpha}}, \quad \left(\frac{Y}{L} \right)^* = A \left(\frac{s}{\delta + n + x} \right)^{\frac{\alpha}{1-\alpha}}.$$

Long-run income depends on technology A and on the parameters that determine steady-state capital intensity.

4.3 Convergence to the steady state

Divide (8) by $\hat{k}$ to get the growth rate of $\hat{k}$:

$$\frac{\dot{\hat{k}}}{\hat{k}} = s\hat{k}^{\alpha-1} - (\delta + n + x), \quad \frac{\partial(\dot{\hat{k}}/\hat{k})}{\partial \hat{k}} = (\alpha - 1)s\hat{k}^{\alpha-2} < 0 \quad (\text{because } \alpha < 1).$$

The growth rate falls steadily as $\hat{k}$ rises: it is very large when $\hat{k}$ is close to 0, it approaches $-(\delta + n + x) < 0$ when $\hat{k}$ is very large, and it equals zero only at $\hat{k}^*$. So it is positive below $\hat{k}^*$ and negative above it.

Global stability

The system converges to the steady state from *any* starting value $\hat{k}_0 > 0$. $\hat{k}$ grows only during the transition; once $\hat{k}^*$ is reached, growth in capital per effective worker stops.

5 Balanced Growth: The Long-Run Facts

Slides: EV 2–6; L1 14

Question (EV slide 2): does the model reproduce the balanced-growth properties of developed economies?

5.1 Output per person grows at the rate of technological progress

Output per person is $y = A\hat{y} = A\hat{k}^\alpha$. By the growth-rate rules:

$$\frac{\dot{y}}{y} = \frac{\dot{A}}{A} + \frac{\dot{\hat{y}}}{\hat{y}} = x + \alpha \frac{\dot{\hat{k}}}{\hat{k}}.$$

As $\hat{k} \rightarrow \hat{k}^*$, $\dot{\hat{k}}/\hat{k} \rightarrow 0$. **Hence** $\dot{y}/y \rightarrow x$.

5.2 Consumption and investment grow with output

The resource constraint and the constant saving rate give $C = (1 - s)Y$ and $I = sY$. Since s is constant,

$$\frac{\dot{C}}{C} = \frac{\dot{Y}}{Y}, \quad \frac{\dot{I}}{I} = \frac{\dot{Y}}{Y}.$$

In the long run output, consumption and investment *per person* all grow at rate x . This broadly describes developed economies; in many developing economies, however, saving rates rise during development.

5.3 Capital grows faster than labour

From $\hat{k} = K/(AL)$:

$$\frac{\dot{K}}{K} = \frac{\dot{A}}{A} + \frac{\dot{L}}{L} + \frac{\dot{\hat{k}}}{\hat{k}} = x + n + \frac{\dot{\hat{k}}}{\hat{k}}.$$

On the balanced growth path $\dot{\hat{k}}/\hat{k} = 0$, so

$$\frac{\dot{K}}{K} = n + x \quad \text{and} \quad \frac{\dot{K}}{K} - \frac{\dot{L}}{L} = x.$$

5.4 The return to capital is constant

The net return to capital is

$$r - \delta = \alpha(AL)^{1-\alpha}K^{\alpha-1} - \delta = \alpha\hat{k}^{\alpha-1} - \delta.$$

On the balanced growth path $\hat{k}$ is constant, so $r - \delta$ is constant. K and AL both grow, but their effects on the marginal product of capital offset each other.

5.5 Summary of long-run growth rates

Variable		Long-run growth rate
Per effective worker	$\hat{k}, \hat{y}, \hat{c}, \hat{i}$	0
Per capita	y, c, i, k , and the wage w	x
Aggregates	Y, C, I, K	$n + x$
Rental rate	r (and $r - \delta$)	0
Ratios	$K/Y, C/Y, I/Y$	0

In one paragraph In the long run everything “per effective worker” is constant, everything “per person” grows at the technology rate x , and aggregates grow at $n + x$. That is the model’s *balanced growth path*.

6 Transitional Growth and Convergence

Slides: EV 7–13

6.1 Long-run growth vs. transitional growth

Combining $\dot{y}/y = x + \alpha \dot{\hat{k}}/\hat{k}$ with the growth rate of $\hat{k}$:

$$\frac{\dot{y}}{y} = \underbrace{x}_{\text{long-run growth}} + \underbrace{\alpha \frac{\dot{\hat{k}}}{\hat{k}}}_{\text{transitional growth}}, \quad \frac{\dot{\hat{k}}}{\hat{k}} = s\hat{k}^{\alpha-1} - \delta - n - x, \quad \frac{\partial(\dot{\hat{k}}/\hat{k})}{\partial \hat{k}} < 0.$$

Key mechanism An economy with little capital per effective worker has a *high marginal product of capital*, so it grows fast during its transition to the steady state. As capital accumulates, diminishing returns set in and the extra growth fades.

6.2 Conditional convergence, no absolute convergence, divergence

- **Conditional convergence.** Suppose a country is poor because its initial K/L is low, while s, n, x, δ and the technology A are the same as elsewhere. Then

$$\hat{k} \text{ low} \implies Y/L \text{ low} \quad \text{and} \quad \dot{\hat{k}}/\hat{k} \text{ high} \implies \dot{y}/y \text{ high.}$$

Countries converge to the same steady state when the parameters that determine it are the same.

- **Why absolute convergence need not occur.** If a country is poor because its technology A is low, its $\hat{k} = K/(AL)$ may be *high*. Then

$$Y/L \text{ low}, \quad \hat{k} \text{ high}, \quad \dot{\hat{k}}/\hat{k} \text{ low}, \quad \dot{y}/y \text{ low.}$$

Differences in s, n or x also produce different steady states. A poor country need not grow faster than a rich one.

- **Divergence.** Take two countries that start with the same income per worker. If one saves more, it converges to a higher steady-state *level*; if one has faster technological progress, its long-run *growth rate* is higher. Their incomes diverge.

Prediction

The Solow model predicts **conditional** convergence, not convergence among all countries regardless of their characteristics. Empirically: convergence within groups of similar countries or regions, persistent or widening gaps across dissimilar ones.

6.3 A numerical example

Let $s = 0.25$, $\alpha = 1/3$, $\delta = 0.05$, $n = 0.01$, $x = 0.02$ (so $\delta + n + x = 0.08$), and start at $\hat{k}_0 = 1.8$. Initial growth of output per worker:

$$\begin{aligned}\frac{\dot{y}}{y} &= x + \alpha \left[s\hat{k}_0^{\alpha-1} - (\delta + n + x) \right] \\ &= 0.02 + \frac{1}{3} \left[0.25 \times 1.8^{-2/3} - 0.08 \right] \\ &= 0.02 + \frac{1}{3} [0.169 - 0.080] \approx 0.050.\end{aligned}$$

plug in
 $1.8^{-2/3} \approx 0.676$

Growth starts near **5%**, but the transitional part fades as capital accumulates:

t	$\hat{k}_t$	$\dot{\hat{k}}_t/\hat{k}_t$	$\dot{y}_t/y_t$	t	$\hat{k}_t$	$\dot{\hat{k}}_t/\hat{k}_t$	$\dot{y}_t/y_t$
0	1.8000	—	—	7	2.8267	0.0491	0.0364
1	1.9601	0.0890	0.0497	8	2.9541	0.0451	0.0350
2	2.1162	0.0796	0.0465	9	3.0764	0.0414	0.0338
3	2.2678	0.0717	0.0439	10	3.1939	0.0382	0.0327
4	2.4149	0.0648	0.0416	11	3.3066	0.0353	0.0318
5	2.5571	0.0589	0.0396	12	3.4145	0.0326	0.0309
6	2.6944	0.0537	0.0379	13	3.5178	0.0303	0.0301
				14	3.6166	0.0281	0.0294

Output growth falls quickly toward the long-run rate of 2%.

Beyond the slides The table can be reproduced with annual steps $\hat{k}_{t+1} = \hat{k}_t + s\hat{k}_t^\alpha - 0.08\hat{k}_t$; e.g. $\hat{k}_1 = 1.8 + 0.25(1.8)^{1/3} - 0.144 = 1.9601$. The columns are $\dot{\hat{k}}_t/\hat{k}_t = (\hat{k}_t - \hat{k}_{t-1})/\hat{k}_{t-1}$ and $\dot{y}_t/y_t = x + \alpha \dot{\hat{k}}_t/\hat{k}_t$.

6.4 Limits of transitional growth

Because $\dot{\hat{k}}/\hat{k} \rightarrow 0$, the model has difficulty sustaining a growth miracle for several decades:

- sustained growth near 5% eventually requires technological progress near 5%;
- a prolonged growth disaster similarly requires technological progress near zero or below zero.

Key result

Capital accumulation can generate large *short-run* growth differences, but technology determines *long-run* growth.

7 Explaining International Income Differences

Slides: EV 14–17

Question (EV slide 2): can plausible differences in saving, population growth and technology explain international income gaps?

Start from steady-state income per worker (Section 4.2):

$$y^* = A \left(\frac{s}{\delta + n + x} \right)^{\frac{\alpha}{1-\alpha}}, \quad \frac{\alpha}{1-\alpha} = \frac{1/3}{2/3} = \frac{1}{2}.$$

Can we get a **tenfold** gap, $y_R/y_P = 10$ (R = rich, P = poor)? Change one thing at a time.

7.1 Saving rates

Same A, n, x, δ, α ; different s . Then

$$\begin{aligned} \frac{y_R}{y_P} &= \left(\frac{s_R}{s_P} \right)^{\frac{\alpha}{1-\alpha}} = 10 && \text{only } s \text{ differs} \\ \frac{s_R}{s_P} &= 10^{(1-\alpha)/\alpha} = 10^2 = 100. && \text{raise to the power } \frac{1-\alpha}{\alpha} = 2 \end{aligned}$$

If $s_R = 0.25$, then $s_P = 0.25/100 = 0.0025$.

Assessment. Poor countries do tend to save less, but a 0.25% saving rate is implausibly low. Saving differences explain only part of the gap.

7.2 Population growth

Same A and s ; different n . Then

$$\begin{aligned} \frac{y_R}{y_P} &= \left(\frac{\delta + n_P + x}{\delta + n_R + x} \right)^{\frac{\alpha}{1-\alpha}} = 10 && \text{only } n \text{ differs} \\ \delta + n_P + x &= 100(\delta + n_R + x) && \text{square both sides} \\ n_P &= 100(\delta + n_R + x) - \delta - x = 100(0.08) - 0.07 = 7.93. && \delta = 0.05, n_R = 0.01, x = 0.02 \end{aligned}$$

That is about **793% per year**.

Assessment. Population-growth differences of plausible size cannot explain a tenfold income gap.

7.3 Technology

All other parameters the same. Then

$$\frac{y_R}{y_P} = \frac{A_R}{A_P} = 10.$$

Assessment. Differences in technology / total factor productivity can, in accounting terms, generate large income differences. *Limitation:* the basic Solow model treats the level of technology as exogenous, so it does not explain why technology differs across countries.

Source of the gap	Required for $y_R/y_P = 10$	Plausible?
Saving rate s	$s_R/s_P = 100$ (e.g. $s_P = 0.25\%$)	No
Population growth n	$n_P \approx 793\%$ per year	No
Technology A	$A_R/A_P = 10$	Yes, but A is unexplained

Intuition Because of diminishing returns to capital, steady-state income responds only weakly to saving and population growth (elasticity $\alpha/(1-\alpha) = 1/2$). Huge differences in these parameters are needed to produce modest income differences.

8 Solving the Solow Differential Equation

Slides: DE 1–14

Roadmap of this section (1) Learn to solve a *linear* first-order ODE. (2) Turn the (nonlinear) Solow equation into a linear one and solve it *exactly*. (3) Solve it *approximately* by log-linearizing. (4) Translate capital convergence into *income* convergence. (5) Turn that into a cross-country *growth regression*. (6) Compute the *speed* of convergence and the half-life.

8.1 Linear first-order differential equations

A first-order differential equation relates a variable, its time derivative, and possibly time: $\dot{z}_t = g(z_t, t)$. A **linear** one has the form

$$\dot{z}_t + a(t)z_t = b(t).$$

Together with an initial condition z_0 , it determines the path z_t . Take constant coefficients $a > 0$ and b :

$$\dot{z}_t + az_t = b.$$

Solving it with an integrating factor

Step 1. Multiply both sides by e^{at} :

$$e^{at}\dot{z}_t + ae^{at}z_t = be^{at}.$$

Step 2. The left-hand side is the derivative of a product (product rule: $\frac{d}{dt}(e^{at}z_t) = e^{at}\dot{z}_t + ae^{at}z_t$):

$$\frac{d}{dt}(e^{at}z_t) = be^{at}.$$

Step 3. Integrate both sides over time (C is an arbitrary constant):

$$\int \frac{d}{dt}(e^{at}z_t) dt = \int be^{at} dt \implies e^{at}z_t = \frac{b}{a}e^{at} + C.$$

Step 4. Divide by e^{at} — the family of all solutions:

$$z_t = \frac{b}{a} + Ce^{-at}.$$

Step 5. Pin down C with the initial condition. At $t = 0$, $e^{-a \cdot 0} = 1$, so

$$z_0 = \frac{b}{a} + C \implies C = z_0 - \frac{b}{a}.$$

Solution of $\dot{z}_t + az_t = b$

$$z_t = \frac{b}{a} + \left(z_0 - \frac{b}{a}\right)e^{-at} \quad \text{or, with } z^* \equiv b/a, \quad z_t = z^* + (z_0 - z^*)e^{-at}.$$

What the solution says

- The gap $z_t - z^*$ shrinks at exponential rate a . Differentiating with respect to time:

$$\frac{d}{dt}(z_t - z^*) = -a(z_t - z^*)e^{-at} = -a(z_t - z^*).$$

- If $z_0 < z^*$ the path rises toward z^* ; if $z_0 > z^*$ it falls.
- At time t , the fraction of the initial gap that has been closed is $1 - e^{-at}$.

8.2 The exact Solow transition path

With Cobb–Douglas, the Solow equation

$$\dot{\hat{k}}_t = s\hat{k}_t^\alpha - (\delta + n + x)\hat{k}_t, \quad 0 < \alpha < 1,$$

is *nonlinear* because of the $\hat{k}_t^\alpha$ term, so the formula above does not apply directly. Its positive steady state is $\hat{k}^* = [s/(\delta + n + x)]^{1/(1-\alpha)}$. It is a **Bernoulli equation**: a change of variable makes it linear.

Step 1. For $\hat{k}_t > 0$, divide the equation by $\hat{k}_t^\alpha$ (after moving the break-even term to the left):

$$\hat{k}_t^{-\alpha}\dot{\hat{k}}_t + (\delta + n + x)\hat{k}_t^{1-\alpha} = s.$$

Step 2. Define the new variable $z_t \equiv \hat{k}_t^{1-\alpha}$. By the chain rule,

$$\dot{z}_t = (1 - \alpha)\hat{k}_t^{-\alpha}\dot{\hat{k}}_t \implies \hat{k}_t^{-\alpha}\dot{\hat{k}}_t = \frac{\dot{z}_t}{1 - \alpha}.$$

Step 3. Substitute into Step 1 and multiply by $(1 - \alpha)$:

$$\dot{z}_t + \underbrace{(1 - \alpha)(\delta + n + x)}_a z_t = \underbrace{(1 - \alpha)s}_b.$$

This is **linear in z_t** , exactly the form we just solved.

Step 4. Its steady state is

$$z^* = \frac{b}{a} = \frac{(1 - \alpha)s}{(1 - \alpha)(\delta + n + x)} = \frac{s}{\delta + n + x} = (\hat{k}^*)^{1-\alpha}.$$

Step 5. Apply the linear solution, then undo the substitution ($\hat{k}_t = z_t^{1/(1-\alpha)}$, $z_0 = \hat{k}_0^{1-\alpha}$).

Exact Solow path

$$z_t = z^* + (z_0 - z^*)e^{-\beta t}, \quad \beta \equiv (1 - \alpha)(\delta + n + x),$$

$$\hat{k}_t = \left[z^* + (\hat{k}_0^{1-\alpha} - z^*) e^{-\beta t} \right]^{\frac{1}{1-\alpha}}.$$

Exact and approximate convergence

The exact solution says that the gap in $z_t = \hat{k}_t^{1-\alpha}$ closes at the rate $\beta = (1 - \alpha)(\delta + n + x)$. The path for $\hat{k}_t$ itself is nonlinear. Near the steady state, however, its *logarithm* approximately follows the same exponential convergence law (next subsection).

Why bother with an approximation? Log output per effective worker is proportional to log capital per effective worker ($\ln \hat{y} = \alpha \ln \hat{k}$), which makes the log version useful for measuring *income* convergence.

Beyond the slides — numerical check With the Section 6.3 numbers ($s = 0.25$, $\alpha = 1/3$, $\delta + n + x = 0.08$, $\hat{k}_0 = 1.8$): $\beta = \frac{2}{3}(0.08) \approx 0.0533$, $z^* = 0.25/0.08 = 3.125$, $z_0 = 1.8^{2/3} \approx 1.4798$. At $t = 14$:

$$z_{14} = 3.125 - 1.6452 e^{-0.0533 \times 14} \approx 3.125 - 1.6452 \times 0.4739 \approx 2.3453, \quad \hat{k}_{14} = 2.3453^{3/2} \approx 3.59,$$

close to the table's 3.617 (the table uses annual steps, not continuous time).

8.3 Log-linearizing around the steady state

Step 1. Work in logs. Let $\tilde{k}_t \equiv \ln \hat{k}_t$, so $\hat{k}_t = e^{\tilde{k}_t}$. Then

$$\begin{aligned} \dot{\hat{k}}_t = \frac{\dot{\hat{k}}_t}{\hat{k}_t} &= s \hat{k}_t^{\alpha-1} - (\delta + n + x) && \text{growth rate of } \hat{k} \\ &= s e^{-(1-\alpha)\tilde{k}_t} - (\delta + n + x) \equiv h(\tilde{k}_t). && \hat{k}^{\alpha-1} = e^{(\alpha-1)\tilde{k}} \end{aligned}$$

Step 2. First-order Taylor expansion of h around $\tilde{k}^* = \ln \hat{k}^*$:

$$\dot{\hat{k}}_t = h(\tilde{k}_t) \simeq \underbrace{h(\tilde{k}^*)}_{=\dot{\hat{k}}^*=0} + h'(\tilde{k}^*)(\tilde{k}_t - \tilde{k}^*).$$

The first term is zero because $\tilde{k}$ does not change in the steady state. So we only need the slope $h'(\tilde{k}^*)$.

Step 3. Differentiate h and evaluate at the steady state:

$$\begin{aligned}
 h'(\tilde{k}) &= -(1-\alpha)se^{-(1-\alpha)\tilde{k}} && \text{chain rule} \\
 se^{-(1-\alpha)\tilde{k}^*} &= \delta + n + x && \text{steady state: } s(\hat{k}^*)^{\alpha-1} = (\delta + n + x) \\
 h'(\tilde{k}^*) &= -(1-\alpha)(\delta + n + x) \equiv -\beta. && \text{combine}
 \end{aligned}$$

Step 4. Substitute into the Taylor approximation:

$$\dot{\tilde{k}}_t \simeq -\beta(\tilde{k}_t - \tilde{k}^*), \quad \beta = (1-\alpha)(\delta + n + x).$$

8.4 Solving the log-linear equation

Rearrange it into the familiar linear form:

$$\dot{\tilde{k}}_t + \beta\tilde{k}_t = \beta\tilde{k}^*.$$

Match it with $\dot{z}_t + az_t = b$:

$$z_t \longleftrightarrow \tilde{k}_t, \quad a \longleftrightarrow \beta, \quad b \longleftrightarrow \beta\tilde{k}^*, \quad \text{so } \frac{b}{a} = \tilde{k}^*.$$

Applying $z_t = b/a + (z_0 - b/a)e^{-at}$:

Log-linear path

$$\tilde{k}_t \simeq \tilde{k}^* + (\tilde{k}_0 - \tilde{k}^*)e^{-\beta t} \quad \Longleftrightarrow \quad \tilde{k}_t - \tilde{k}_0 \simeq (1 - e^{-\beta t})(\tilde{k}^* - \tilde{k}_0).$$

Log capital closes its gap to the steady state at (approximately) rate β per year.

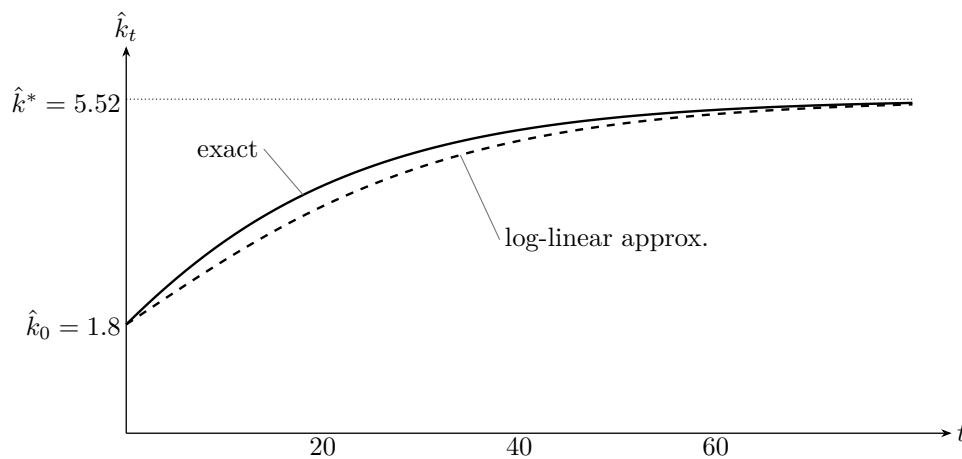


Figure 2: Exact path vs. log-linear approximation ($\hat{k}_0 = 1.8$, $\hat{k}^* = 5.52$, $\beta = 0.0533$). Far from the steady state the approximation understates the early catch-up; the two paths merge as $\hat{k}_t \rightarrow \hat{k}^*$. (Illustration, not on the slides.)

8.5 From capital convergence to income convergence

Let $y_t \equiv Y_t/L_t = A_t \hat{y}_t$ with $\hat{y}_t = \hat{k}_t^\alpha$.

$$\begin{aligned} \ln y_t &= \ln A_t + \alpha \tilde{k}_t, & \ln \hat{y}^* &= \alpha \tilde{k}^* & \text{take logs; } \tilde{k} = \ln \hat{k} \\ \ln y_t - \ln y_0 &= (\ln A_t - \ln A_0) + \alpha(\tilde{k}_t - \tilde{k}_0) & & \text{subtract the } t=0 \text{ version} \\ &= xt + \alpha(\tilde{k}_t - \tilde{k}_0) & & A \text{ grows at rate } x \\ &\simeq xt + (1 - e^{-\beta t}) \alpha(\tilde{k}^* - \tilde{k}_0). & & \text{log-linear path} \end{aligned}$$

Finally, $\alpha \tilde{k}^* = \ln \hat{y}^*$ and $\alpha \tilde{k}_0 = \ln \hat{y}_0 = \ln y_0 - \ln A_0$, so

Income convergence

$$\ln y_t - \ln y_0 \simeq xt + (1 - e^{-\beta t}) [\ln \hat{y}^* - (\ln y_0 - \ln A_0)].$$

Holding fixed the determinants of $\hat{y}^*$ (and A_0), a **lower initial income** y_0 **predicts faster transitional growth**. The approximation applies near the steady state.

8.6 From the Solow transition to a growth regression

Let i index countries and T be the length of the observation interval.

Step 1. Write the income-convergence equation for country i at $t = T$ and divide by T :

$$\frac{\ln y_{i,T} - \ln y_{i,0}}{T} \simeq x + \frac{1 - e^{-\beta T}}{T} [\ln \hat{y}_i^* - \ln y_{i,0} + \ln A_{i,0}].$$

Step 2. Define the convergence coefficient $\lambda_T \equiv \frac{1 - e^{-\beta T}}{T}$.

Step 3. Substitute the Cobb–Douglas steady state:

$$\ln \hat{y}_i^* = \frac{\alpha}{1 - \alpha} \ln s_i - \frac{\alpha}{1 - \alpha} \ln(n_i + x + \delta).$$

Step 4. Collect terms:

Growth regression

$$\frac{\ln y_{i,T} - \ln y_{i,0}}{T} \simeq c_i - \lambda_T \ln y_{i,0} + \lambda_T \frac{\alpha}{1 - \alpha} \ln s_i - \lambda_T \frac{\alpha}{1 - \alpha} \ln(n_i + x + \delta), \quad c_i \equiv x + \lambda_T \ln A_{i,0}.$$

For estimation, initial productivity may be related to observed country characteristics Z_i (e.g. rule of law, life expectancy, inflation):

$$c_i = c_0 + \gamma' Z_i + u_i.$$

Conditional convergence predicts a negative coefficient on initial income ($-\lambda_T < 0$), holding the determinants of each country's steady state fixed.

8.7 Speed of convergence and half-life

In the basic model $\beta = (1 - \alpha)(\delta + n + x)$. With $\alpha = 1/3$, $\delta = 0.05$, $n = 0.01$, $x = 0.02$ per year:

$$\beta = \frac{2}{3}(0.08) \simeq 0.0533 \text{ per year.}$$

The **half-life** $T_{1/2}$ is the time needed to close half of the initial log gap:

$$e^{-\beta T_{1/2}} = \frac{1}{2} \quad \text{half the gap remains}$$

$$T_{1/2} = -\frac{\ln(1/2)}{\beta} = \frac{\ln 2}{\beta}. \quad \text{take logs}$$

Half-life

model:	$\beta = 0.0533$	$\implies$	$T_{1/2} = 0.693/0.0533 \approx 13 \text{ years,}$
illustrative empirical rate:	$\beta = 0.02$	$\implies$	$T_{1/2} = 0.693/0.02 \approx 35 \text{ years.}$

The model's convergence is too fast compared with the data. The slower empirical rate motivates richer models of convergence.

9 Policy Analysis

Slides: PO 1–8

Three questions (PO slide 1): What saving rate maximizes steady-state consumption? How does more saving affect consumption along the transition? Which forms of foreign aid change income temporarily or permanently? Throughout we use

$$\hat{y} = \hat{k}^\alpha, \quad \hat{c} = (1 - s)\hat{y}, \quad \dot{\hat{k}} = s\hat{k}^\alpha - (\delta + n + x)\hat{k}, \quad 0 < \alpha < 1.$$

9.1 The Golden Rule

In a steady state, investment just covers break-even investment, $\hat{i}^* = (\delta + n + x)\hat{k}^*$, so

$$\hat{c}^* = \hat{y}^* - \hat{i}^* = (\hat{k}^*)^\alpha - (\delta + n + x)\hat{k}^*. \quad \text{consumption} = \text{output} - \text{investment}$$

$$\frac{d\hat{c}^*}{d\hat{k}^*} = \alpha(\hat{k}^*)^{\alpha-1} - (\delta + n + x) = 0 \quad \text{maximize over } \hat{k}^*$$

$$\hat{k}_{GR} = \left(\frac{\alpha}{\delta + n + x} \right)^{\frac{1}{1-\alpha}}. \quad \text{solve}$$

Which saving rate implements the Golden Rule?

Put the two formulas side by side:

$$\hat{k}^*(s) = \left(\frac{s}{\delta + n + x} \right)^{\frac{1}{1-\alpha}} \quad \text{vs.} \quad \hat{k}_{GR} = \left(\frac{\alpha}{\delta + n + x} \right)^{\frac{1}{1-\alpha}}.$$

They have the same form, with s in place of α .

Golden Rule saving rate

$$s_{GR} = \alpha.$$

Under Cobb–Douglas, the saving rate that maximizes steady-state consumption equals capital's income share.

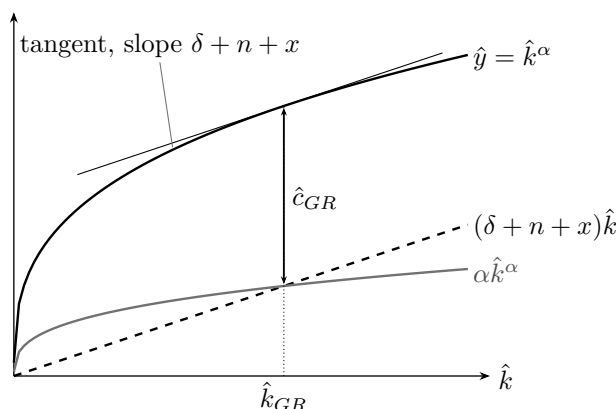


Figure 3: The Golden Rule ($\alpha = 1/3$, $\delta + n + x = 0.08$). Steady-state consumption is the gap between $\hat{y}$ and the break-even line; it is largest where the slope of $\hat{y} = \hat{k}^\alpha$ equals $\delta + n + x$. With $s = \alpha$ the investment curve $\alpha \hat{k}^\alpha$ crosses the break-even line exactly at $\hat{k}_{GR}$.

Beyond the slides — the Golden Rule in words The first-order condition says $f'(\hat{k}_{GR}) = \alpha \hat{k}_{GR}^{\alpha-1} = \delta + n + x$, i.e. $r - \delta = n + x$: at the Golden Rule, the net marginal product of capital equals the growth rate of the economy. The second derivative $\alpha(\alpha - 1)(\hat{k}^*)^{\alpha-2} < 0$ confirms this is a maximum.

9.2 Saving above or below the Golden Rule

$s > \alpha$: overaccumulation

$\hat{k}^* > \hat{k}_{GR}$. Lowering s raises consumption *immediately and* in the new steady state. The economy is **dynamically inefficient**.

$s < \alpha$: underaccumulation

$\hat{k}^* < \hat{k}_{GR}$. Raising s lowers consumption *initially* but raises it *eventually*. Whether that improves welfare depends on how future consumption is valued.

In both cases $\hat{c}^*(\hat{k}) = \hat{k}^\alpha - (\delta + n + x)\hat{k}$ reaches its maximum at $\hat{k}_{GR}$.

9.3 A higher saving rate: the transition

At date t_0 , s rises permanently from s_0 to s_1 . $\hat{k}$ is predetermined (the capital stock cannot jump).

1. Investment $s\hat{y}$ rises at once, so $\dot{\hat{k}}$ becomes positive.
2. $\hat{k}$ and $\hat{y}$ rise gradually toward their higher steady-state levels.

3. Growth of Y/L temporarily exceeds x , then returns to x .
4. $\hat{c} = (1 - s)\hat{y}$ falls on impact; its eventual level may rise or fall, depending on the initial and new saving rates.

Level vs. growth

A higher saving rate raises the *level* of output per worker, but does not permanently raise its *growth rate* in the Solow model.

Beyond the slides — size of the jump Starting from the old steady state, $s_0(\hat{k}_0^*)^{\alpha-1} = (\delta + n + x)$, so right after the change

$$\left. \frac{\dot{\hat{k}}}{\hat{k}} \right|_{t_0} = s_1(\hat{k}_0^*)^{\alpha-1} - (\delta + n + x) = \left(\frac{s_1}{s_0} - 1 \right) (\delta + n + x), \quad \left. \frac{\dot{y}}{y} \right|_{t_0} = x + \alpha \left(\frac{s_1}{s_0} - 1 \right) (\delta + n + x).$$

With $s_0 = 0.2$, $s_1 = 0.3$, $\alpha = 1/3$, $(\delta + n + x) = 0.08$: growth of Y/L jumps from 2% to about 3.3% and then decays back to 2% (figure below). Since $s_1 = 0.3 < \alpha$, consumption ends up higher than before.

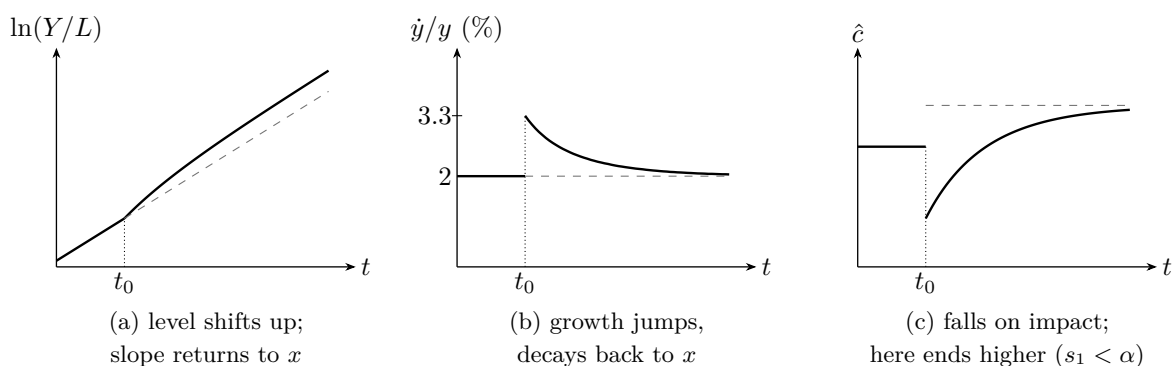


Figure 4: Permanent increase in s from 0.20 to 0.30 at date t_0 ($\alpha = 1/3$, $\delta + n + x = 0.08$, $x = 0.02$). (Illustration, not on the slides.)

9.4 Foreign aid

Consumption goods

A temporary transfer raises consumption while it lasts. If it leaves saving, capital and productivity unchanged, it leaves the economy's **production path unchanged**.

Capital goods or investment finance

If aid finances productive investment while $\hat{k}$ is below its steady state, it speeds up capital accumulation during the transfer. After temporary aid ends, capital converges back toward the **original** steady state — unless the transfer changes domestic saving or productivity persistently.

How a temporary investment transfer works

Let a_t be aid per effective worker, with a fraction φ invested directly. If the remaining share $(1 - \varphi)$ is treated as income and saved at rate s :

$$\dot{\hat{k}}_t = s \hat{k}_t^\alpha + \underbrace{[\varphi + s(1 - \varphi)] a_t}_{\text{investment financed by aid}} - (\delta + n + x) \hat{k}_t.$$

- Cash aid entirely treated as disposable income: $\varphi = 0$, extra investment sa_t .
- In-kind investment aid: $\varphi = 1$, extra investment a_t .
- If $a_t = 0$ after a finite period, the equation returns to its original form, and the lasting effect on $\hat{k}$ vanishes as the economy approaches its original steady state.

9.5 Technical assistance and the level of technology

If assistance permanently raises labour-augmenting productivity A_t , output per worker $Y_t/L_t = A_t f(K_t/(A_t L_t))$ can be permanently higher. At the instant A_t rises, physical capital K_t is fixed, so $\hat{k}_t = K_t/(A_t L_t)$ falls. Capital then accumulates back toward the original steady-state value of $\hat{k}$ (if s, n, x, δ are unchanged).

Level vs. growth, again

A permanent one-time increase in the *level* of A_t raises the long-run level of output per worker.
A permanently higher *growth rate* requires a persistent increase in the growth rate x itself.

Beyond the slides — a worked version Let A rise to μA ($\mu > 1$) at t_0 , starting from $\hat{k}^*$. Then $\hat{k}$ drops to $\hat{k}^*/\mu$ and output per worker moves as

$$A(\hat{k}^*)^\alpha \xrightarrow{\text{impact}} \mu A \left(\frac{\hat{k}^*}{\mu} \right)^\alpha = \mu^{1-\alpha} A(\hat{k}^*)^\alpha \xrightarrow{\text{long run}} \mu A(\hat{k}^*)^\alpha :$$

an immediate jump by the factor $\mu^{1-\alpha}$, then capital accumulation takes it the rest of the way to μ times the original level.

10 Overall Assessment and Limitations

Slides: L1 14–15; EV 18

10.1 What the model gets right

Do the variables in the model behave like those in the data? The model is consistent with several long-run facts (L1 slide 14):

1. Output per capita grows at a constant rate in the long run.
2. C/Y and I/Y are constant in the long run.
3. Capital grows at the constant rate x faster than population in the long run.
4. The rate of return to capital has no long-run trend.

5. It can generate large and persistent differences in income levels and growth rates through differences in s , A and x .
- 6–7. It predicts conditional, but not absolute, convergence.

10.2 Overall assessment

- It reproduces key balanced-growth facts for developed economies.
- It predicts conditional convergence through diminishing returns to capital.
- Capital accumulation can generate temporarily high growth, but not permanently high growth.
- Plausible saving and population-growth differences explain only part of international income gaps.
- Technology can account for large income differences, but the model leaves technology exogenous.

10.3 Limitations of the baseline model

1. **Key components are exogenous.** The saving rate, productivity and technological change are all exogenous, so the explanation is incomplete, and the model is not well suited to policy analysis.
2. **Expenditure ratios.** C/Y and I/Y are also constant during the transitional growth period, which is counterfactual.
3. **Convergence is too fast.** The implied speed of convergence is too high ($\beta \approx 5.3\%$, half-life ≈ 13 years, vs. roughly 2% and 35 years in the data), motivating extensions of the basic framework.

Central lesson

The Solow model organizes the *proximate* sources of growth and income differences (capital accumulation and technology), while leaving their deeper causes outside the model.

Formula Sheet

Growth-rate rules	$\frac{(\dot{XY})}{XY} = \frac{\dot{X}}{X} + \frac{\dot{Y}}{Y}, \quad \frac{(X/Y)}{X/Y} = \frac{\dot{X}}{X} - \frac{\dot{Y}}{Y}, \quad \frac{(\dot{X}^a)}{X^a} = a \frac{\dot{X}}{X}$
Exogenous growth	$L_t = L_0 e^{nt}, \quad A_t = A_0 e^{xt}, \quad A_t L_t \text{ grows at } n + x$
Production	$Y = (AL)^{1-\alpha} K^\alpha, \quad y = A^{1-\alpha} k^\alpha, \quad \hat{y} = \hat{k}^\alpha$
Factor prices	$r = \alpha \hat{k}^{\alpha-1}, \quad w = (1-\alpha) A \hat{k}^\alpha, \quad \text{shares } \alpha \text{ and } 1-\alpha$
Consumption	$\hat{c} = (1-s)\hat{y}$
Solow equation	$\dot{\hat{k}} = s\hat{k}^\alpha - (\delta + n + x)\hat{k}$
Steady state	$\hat{k}^* = \left(\frac{s}{\delta + n + x} \right)^{\frac{1}{1-\alpha}}, \quad y^* = A \left(\frac{s}{\delta + n + x} \right)^{\frac{\alpha}{1-\alpha}}$
Growth decomposition	$\frac{\dot{y}}{y} = x + \alpha \frac{\dot{\hat{k}}}{\hat{k}}, \quad \frac{\dot{\hat{k}}}{\hat{k}} = s\hat{k}^{\alpha-1} - (\delta + n + x)$
Long run	per capita grows at x ; aggregates at $n + x$; $r, K/Y, C/Y, I/Y$ constant
Linear ODE	$\dot{z} + az = b \Rightarrow z_t = z^* + (z_0 - z^*)e^{-at}, \quad z^* = b/a$
Exact Solow path	$\hat{k}_t = \left[z^* + (\hat{k}_0^{1-\alpha} - z^*)e^{-\beta t} \right]^{\frac{1}{1-\alpha}}, \quad z^* = \frac{s}{\delta + n + x}$
Speed of convergence	$\beta = (1-\alpha)(\delta + n + x)$
Log-linear path	$\tilde{k}_t - \tilde{k}_0 \simeq (1 - e^{-\beta t})(\tilde{k}^* - \tilde{k}_0), \quad \tilde{k} = \ln \hat{k}$
Income convergence	$\ln y_t - \ln y_0 \simeq xt + (1 - e^{-\beta t}) [\ln \hat{y}^* - (\ln y_0 - \ln A_0)]$
Growth regression	$\frac{\ln y_{i,T} - \ln y_{i,0}}{T} \simeq c_i - \lambda_T \ln y_{i,0} + \lambda_T \frac{\alpha}{1-\alpha} \ln s_i - \lambda_T \frac{\alpha}{1-\alpha} \ln(n_i + x + \delta), \quad \lambda_T = \frac{1-e^{-\beta T}}{T}$
Half-life	$T_{1/2} = \ln 2 / \beta \quad (\approx 13 \text{ years at } \beta = 0.0533; \approx 35 \text{ at } 0.02)$
Golden Rule	$\hat{k}_{GR} = \left(\frac{\alpha}{\delta + n + x} \right)^{\frac{1}{1-\alpha}}, \quad s_{GR} = \alpha$
Investment aid	$\dot{\hat{k}} = s\hat{k}^\alpha + [\varphi + s(1-\varphi)]a - (\delta + n + x)\hat{k}$

Source notes

- **Sources:** `solow_growth_model_slides.pdf` (L1, 15 slides), `solow_growth_model_evaluation.pdf` (EV, 18), `solow_differential_equations.pdf` (DE, 14), `solow_policy_slides.pdf` (PO, 8).
- **2026-09-30 readability rewrite:** sections reordered to follow the decks (L1 $\rightarrow$ EV $\rightarrow$ DE $\rightarrow$ PO); every derivation step shown with its reason; key results boxed; material not on the slides moved into dashed “Beyond the slides” boxes; formula sheet added.
- **Corrections vs. previous version:** in the growth-regression derivation the first term is $xt/T = x$ (was printed as xt/T); a stray Markdown **bold** fixed; “Kaldor facts” label and “inputs are essential” removed

(not on the slides) in favour of the slides' own list and wording; limitations now follow L1 slide 15 (three items).

- **Slide typos not copied:** EV slide 6 writes $\alpha \hat{K}^{\alpha-1}$ (should be $\alpha \hat{k}^{\alpha-1}$); L1 slide 15 says “contacts” (“constant”).